

MATH 2211, SPRING 2026: MIDTERM 1 STUDY GUIDE

The exam will consist of 5 sections, the first of which will consist of true or false questions. For this, you will be expected to not only give the right response, but also a correct justification for that response, in one or two sentences.

Do not spend more than 8–9 minutes per problem before moving on to the next. This way you can be sure to at least *attempt* every problem in a substantial way.

As mentioned during Lecture 13, you will be expected to know at least the following topics:

- (1) Proofs by induction (usual and strong): This involves clearly setting up the base case, the inductive hypothesis, and then the inductive step, with the use of the inductive hypothesis also made in comprehensible fashion. See Lecture 2 and Homework 1 for examples.
- (2) Modular arithmetic: Lecture 4
- (3) Fields and their properties:
 - (a) Definition: Lecture 4 and Lecture 5.
 - (b) Examples of various fields: $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{Z}/p\mathbb{Z}$ (p prime), $\mathbb{C}$: Lecture 5.
 - (c) Arithmetic in the complex numbers: Modulus/absolute value, Euler's notation, multiplication using Euler's notation. See Lecture 5 and Lecture 6.
 - (d) Basic properties of fields: Problem 3 of Homework 2.
- (4) Linear systems of equations over fields using row operations and their applications: This appears everywhere, including in Lecture 3, Lecture 6 (finding polynomials with certain properties), Lecture 10 (understanding if one vector is in the span of some other vectors).
- (5) Vector spaces and their properties:
 - (a) Definition: Lecture 6
 - (b) Examples: Lecture 7. Note in particular, the vector spaces F^n , $\text{Poly}(F)$, $\text{Poly}_{\leq m}(F)$, $M_{n \times m}(F)$, the last of which was introduced in Lecture 8.
 - (c) Basic properties: Lecture 7.
 - (d) Subspaces: Lecture 7
- (6) Span of subsets $X \subset V$: Lecture 8
 - (a) Definition.
 - (b) Linear combinations.
 - (c) How to decide if a vector is in the span of another set of vectors: see also Example in Lecture 9.
- (7) Linear independence: Definition and properties. See Lecture 9.
- (8) Bases: Lecture 10.
 - (a) Definition.
 - (b) The standard basis for F^n .
 - (c) Examples.
- (9) The Steinitz Exchange Lemma and its consequences:
 - (a) Statement of the lemma: Lecture 11
 - (b) Corollary 1 from Lecture 11: This explains what it means to be finite dimensional and also the well-definedness of the notion of dimension. You should understand the proof and how the Exchange Lemma is used.
 - (c) Proof of the lemma: Lecture 12
- (10) Finite dimensional spaces and dimension: Lecture 12.
 - (a) Definitions
 - (b) Examples
 - (c) Make note of Observations 1, 2, 3 here, which establish useful properties of bases that we will return to often.